

Batch 2023

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Short answer Questions on

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Computer vision is a subset of artificial intelligence that enables acquire, process, analyse the images and videos to make meaningful decisions similar to human beings.

- a) — Image acquisition and processing
— To enhance the images clearly
— scene understanding
— To make decisions on visual data
- b) — autonomous cars and vehicles
— Health care departments (X-ray, CT scans etc)

②

Computer vision key features are noise removal, image segmentation, Enhancing the images to extract meaningful information from the images and videos etc.

- a) Feature extraction is process to identify important characteristics such as edges, corners, textures and shapes.
- b) object recognition identifies and classifies object in images, enabling applications like surveillance, medical diagnosis and autonomous driving.

there are three different level of image processing

- (i) Low-level image processing
- (ii) Mid-level image processing
- (iii) High level image processing

a) low level image processing involves image enhancement, noise removal, filtering without understanding image content.

b) Mid level

- (i) mid level is for image segmentation.
- (ii) Such as to identify edges, corners and sharpening the image.

High level

- (i) High level is for interpret the image.
- (ii) To make the decisions based on image objects.

A digital image represented as matrix of pixels. Each pixel stores intensity or colour.

a) 'Pixel' is the smallest unit of a digital image that stores brightness or colour.

b) Higher resolution means more pixels resulting in better image clarity and lower resolution reduces image quality.

⑤ Image sensing captures light from a scene using sensors and converts into electrical signals.

a) Image sensor converts light into electrical signals that represents image.

b) - digital camera
- web cam.

⑥ a) Sampling and Quantization convert a continuous image into discrete image.

b) Sampling is the Quantization. is used to analyse sampled images and allows to storage of images.

⑦ Image formation model is about the process of capturing images by using light converting to electrical signals to form an object.

a) Perspective projection is process of projecting 3D image into 2D image plane while preserving depth perception.

b) Illumination provides light that determines the brightness, colour and visibility of an object.

⑤ Image processing improves image clarity, whereas computer vision understands the image and makes decisions.

a) Image processing converts the image into pixels to enhance the image.

b) Image processing

① Enhances the image quality

② It converts into pixels without any understanding

Computer vision

① Understands the image or scene

② produces meaningful information

⑨ a) Computer vision widely used in healthcare to analyse images (x-rays, CT-scans, MRI) to detect diseases.

b) It detects roads, traffic signs, other vehicles to enable safe self driving.

⑩ Digital image fundamentals provide the basis of image analysis, recognition and interpretation.

a) Pixel stores the intensity and colour values together form the complete image

b) Sampling determines the number of pixels, while resolution determines image quality. Higher resolution clarity image, lower resolution - low clarity